Lista - Função do 2º grau - Matemática Elementar

12. Determine as raízes, as coordenadas do vértice, a imagem, faça um estudo do sinal da função.

1a. $f(x) = x^2 - 9$

$a = 1 \quad b = 0 \quad c = -9$

$\Delta = b^2 - 4ac$

$\Delta = 0^2 - 4.1(-9)$

$\Delta = 0 + 36$

$\Delta = \sqrt{36}$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-0 \pm \sqrt{36}}{2.1} = \frac{0 \pm 6}{2}$

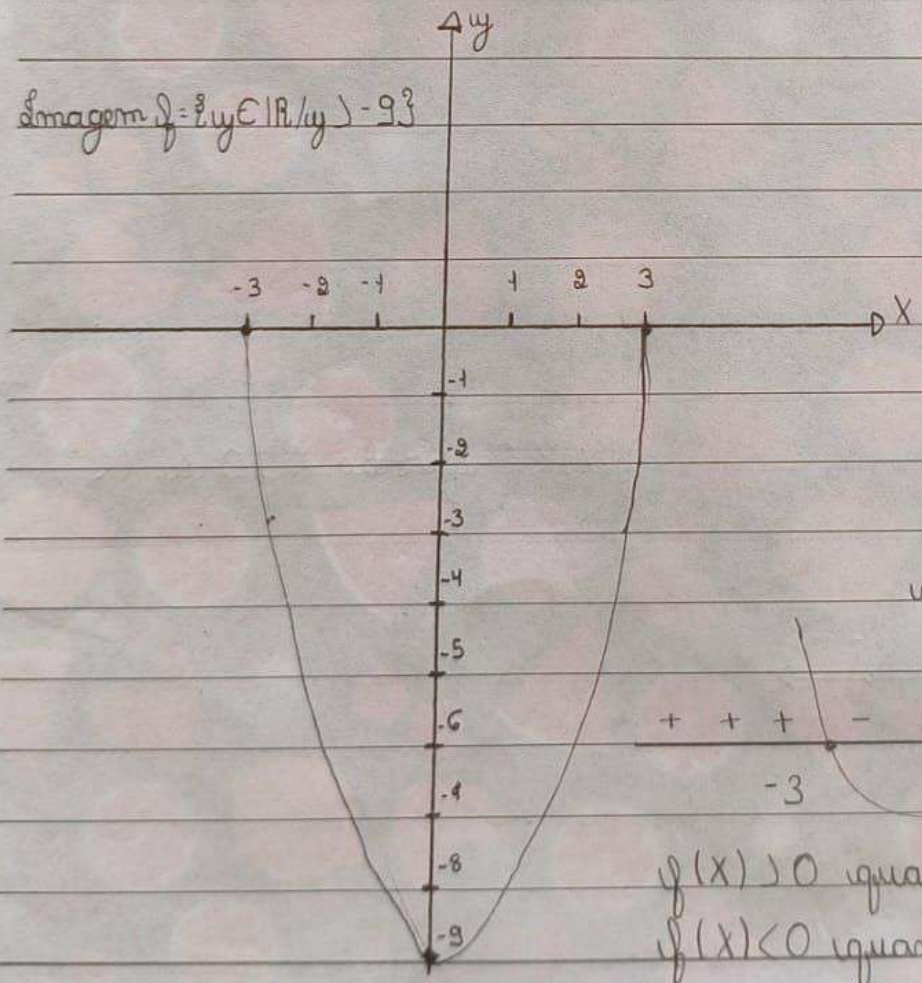
$x' = \frac{0+6}{2} = \frac{6}{2} = 3$

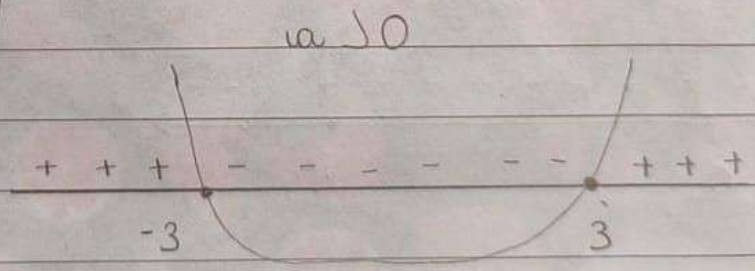
$x'' = \frac{0-6}{2} = \frac{-6}{2} = -3$

$x_v = \frac{-b}{2a} = \frac{-0}{2.1} = \frac{-0}{2} = 0$

$y_v = \frac{-\Delta}{4a} = \frac{-36}{4.1} = \frac{-36}{4} = -9$

Imagem $f = \{y \in \mathbb{R} / y \geq -9\}$





$f(x) > 0$ quando $x < -3$ ou $x > 3$

$f(x) < 0$ quando $-3 < x < 3$

1b. $f(x) = -x^2 + 4x$

$a = -1 \quad b = 4 \quad c = 0$

$\Delta = b^2 - 4 . a . c$

$\Delta = 4^2 - 4 . (-1) . 0$

$\Delta = 16$

$X = \frac{-b \pm \sqrt{\Delta}}{2.a} = \frac{-4 \pm \sqrt{16}}{2.(-1)} = \frac{-4 \pm 4}{-2}$

$\rightarrow X' = \frac{-4+4}{-2} = \frac{0}{-2} = 0$

$\rightarrow X'' = \frac{-4-4}{-2} = \frac{-8}{-2} = 4$

$X_V = \frac{-b}{2.a} = \frac{-4}{2(-1)} = \frac{-4}{-2} = 2$

$y_V = \frac{-\Delta}{4.a} = \frac{-16}{4(-1)} = \frac{-16}{-4} = 4$

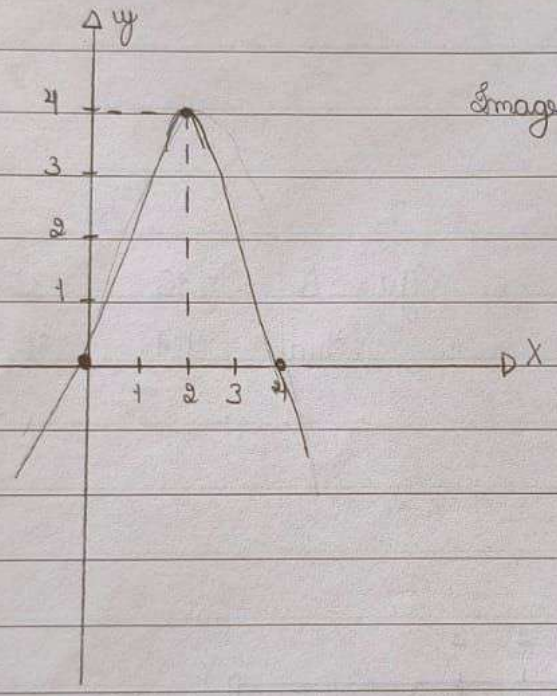
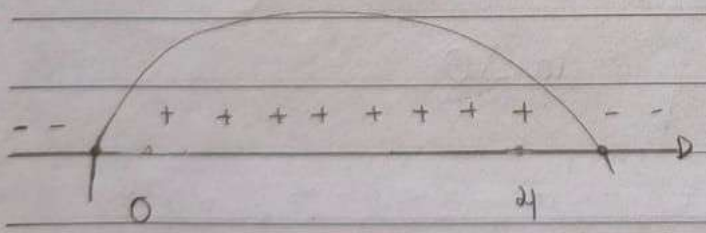


Imagem $f = \{y \in \mathbb{R} / y < 4\}$

$a < 0$

$f(x) < 0$ quando $X > 0$ ou $X < 4$



1c $f(x) = x^2 - 3x + 2$

$a = 1 \quad b = -3 \quad c = 2$

$\Delta = b^2 - 4ac$

$\Delta = -3^2 - 4.1.2$

$\Delta = 9 - 8$

$\Delta = 1$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-(-3) \pm \sqrt{1}}{2.1} = \frac{3 \pm 1}{2}$

$x' = \frac{3+1}{2} = \frac{4}{2} = 2$

$x'' = \frac{3-1}{2} = \frac{2}{2} = 1$

$x_v = \frac{-b}{2a} = \frac{-(-3)}{2.1} = \frac{3}{2} = 1,5$

$y_v = \frac{-\Delta}{4a} = \frac{-1}{4.1} = \frac{-1}{4} = -0,25$

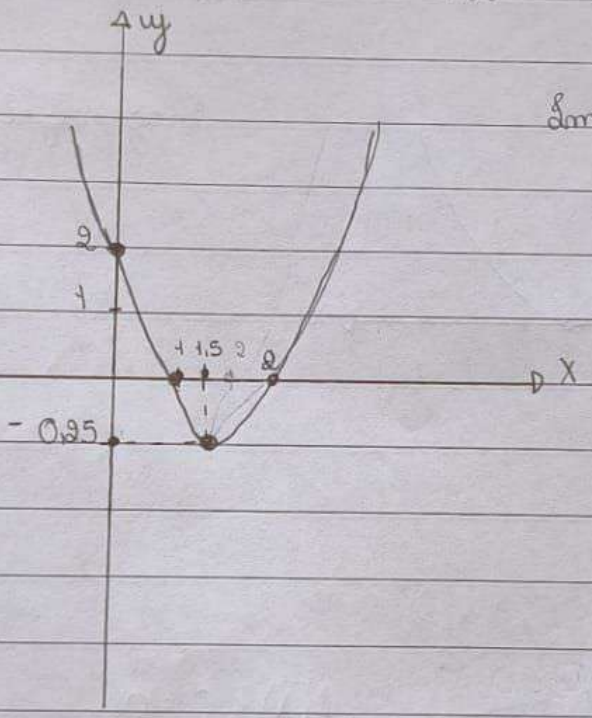
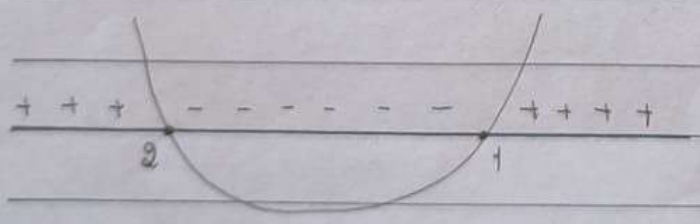


Imagem $\{y \in \mathbb{R} / y \geq -0,25\}$

1a > 0

$f(x) > 0$ quando $x < 2$ ou $x > 1$



1d. $f(x) = x^2 - 4x + 4$

$a = 1 \quad b = -4 \quad c = 4$

$\Delta = b^2 - 4ac$

$\Delta = -4^2 - 4 \cdot 1 \cdot 4$

$\Delta = 16 - 16$

$\Delta = 0$

$x = \frac{-b \pm 0}{2a} = \frac{-(-4) \pm 0}{2 \cdot 1} = \frac{4 \pm 0}{2} = \frac{4}{2} = 2$

$x_v = \frac{-b}{2a} = \frac{-(-4)}{2 \cdot 1} = \frac{4}{2} = 2 \qquad y_v = \frac{-\Delta}{4a} = \frac{0}{4 \cdot 1} = \frac{0}{4} = 0$

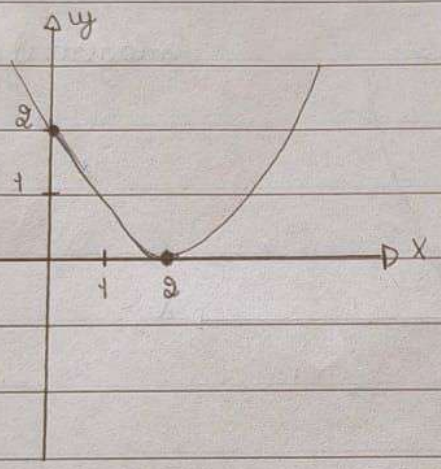
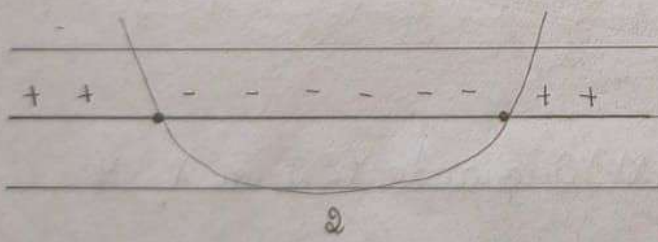


Imagem $f = \{y \in \mathbb{R} / y < 2\}$

a) 0



$f(x) < 0 = \emptyset$

1e. $f(x) = x^2 - x - 6$

$a = 1 \quad b = -1 \quad c = -6$

$\Delta = b^2 - 4ac$

$\Delta = (-1)^2 - 4 \cdot 1 \cdot -6$

$\Delta = 1 + 24$

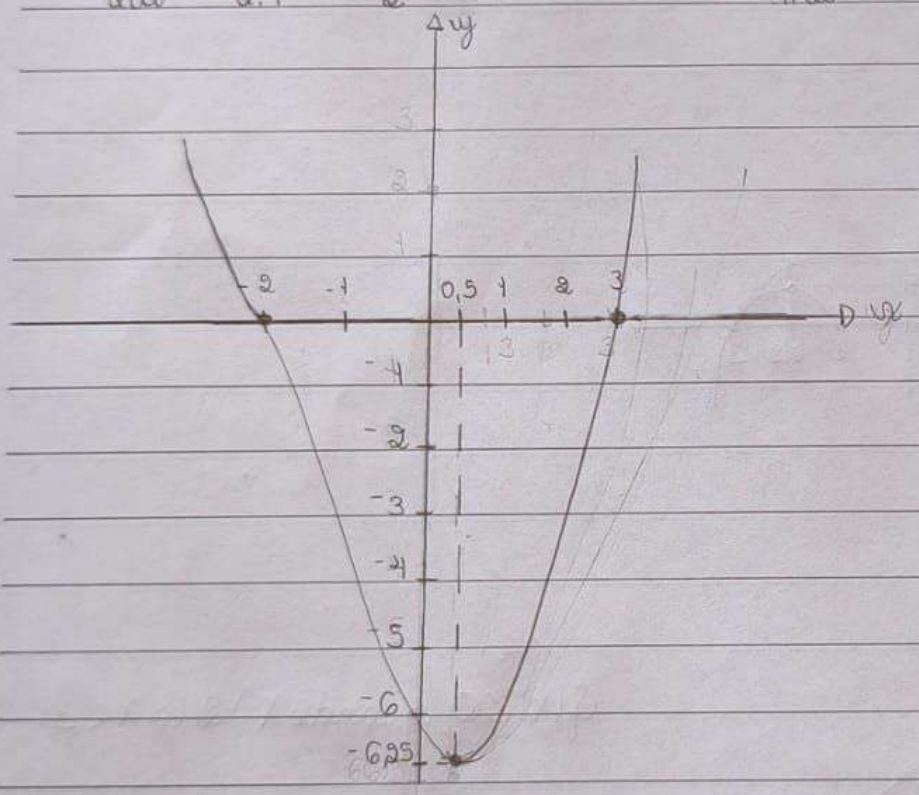
$\Delta = 25$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-(-1) \pm \sqrt{25}}{2 \cdot 1} = \frac{1 \pm 5}{2}$

$x' = \frac{1+5}{2} = \frac{6}{2} = 3$

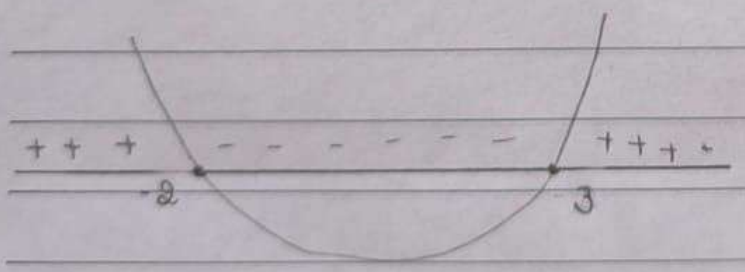
$x'' = \frac{1-5}{2} = \frac{-4}{2} = -2$

$x_v = \frac{-b}{2a} = \frac{-(-1)}{2 \cdot 1} = \frac{1}{2} = 0,5 \qquad x_y = \frac{-\Delta}{4a} = \frac{-25}{4 \cdot 1} = \frac{-25}{4} = -6,25$



$f(x) > 0$ quando $x < -2$ ou $x > 3$

$a > 0$



if. $f(x) = -x^2 + 5x - 6$

$a = -1 \quad b = 5 \quad c = -6$

$\Delta = b^2 - 4ac$

$\Delta = 5^2 - 4 \cdot -1 \cdot -6$

$\Delta = 25 - 24$

$\Delta = 1$

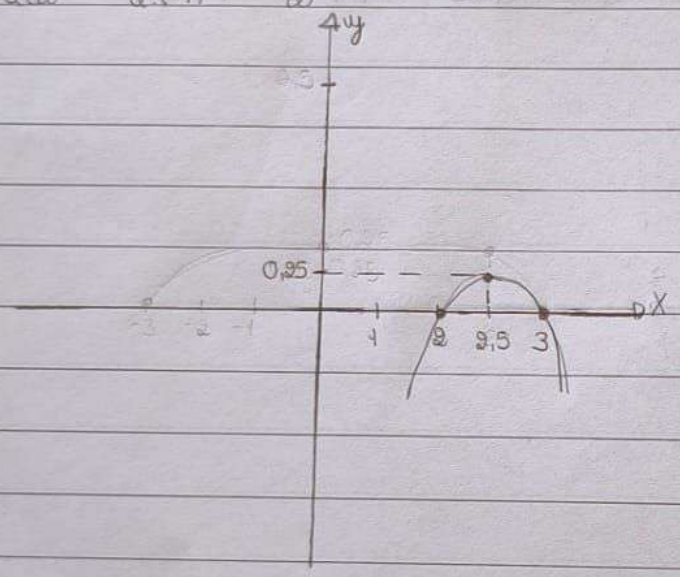
$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-5 \pm \sqrt{1}}{2 \cdot -1} = \frac{-5 \pm 1}{-2}$

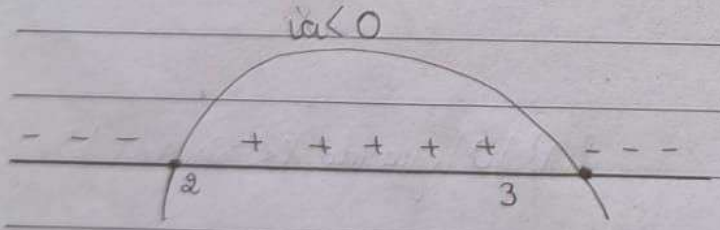
$\rightarrow x' = \frac{-5+1}{-2} = \frac{-4}{-2} = 2$

$\rightarrow x'' = \frac{-5-1}{-2} = \frac{-6}{-2} = 3$

$x_v = \frac{-b}{2a} = \frac{-5}{2(-1)} = \frac{-5}{-2} = 2,5$

$y_v = \frac{-\Delta}{4a} = \frac{-1}{4 \cdot -1} = \frac{-1}{-4} = 0,25$





$f(x) < 0$ quando $x > 2$ e $x < 3$